Implement Iris Flower Classification using Naive Bayes classifier

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.datasets import load_iris
iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
y_pred = nb_model.predict(X_test)
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Accuracy Score:", accuracy_score(y_test, y_pred))
```

```
Confusion Matrix:
 [[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
Accuracy Score: 0.9777777777777777
```

Implementation of Naive Bayes classification for Bank Loan prediction

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score
data = {
    'Income': [40000, 30000, 50000, 60000, 25000, 80000],
    'Age': [25, 35, 45, 40, 22, 50],
    'LoanAmount': [10000, 8000, 15000, 20000, 5000, 25000],
    'CreditScore': [700, 650, 720, 680, 600, 750],
    'Loan_Status': [1, 0, 1, 1, 0, 1]  # Target variable
}
df = pd.DataFrame(data)
X = df[['Income', 'Age', 'LoanAmount', 'CreditScore']]
y = df['Loan_Status']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
y_pred = nb_model.predict(X_test)
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Accuracy Score:", accuracy_score(y_test, y_pred))
```

```
Confusion Matrix:
 [[0 1]
 [0 1]]
Accuracy Score: 0.5
```